

An High-Speed Low-Noise Rotor Topology for EV/HEV PMSM

Hongliang Ying, Surong Huang and Dong Xu

Abstract—Permanent magnet synchronous motor (PMSM) for EV/HEV pursues higher efficiency, higher power density, higher rotation speed and better NVH performance. To meet these requirements, an improved triangle rotor topology is presented. The new rotor topology is researched through comparing the performance with V shape rotor and traditional triangle shape rotor. The comparing results on air-gap flux density, key order radial forces, sound pressure, inductance, torque, anti-demagnetization capability and efficiency proved the advantage of the new rotor topology.

Index Terms—High speed, PMSM, noise, rotor topology.

I. INTRODUCTION

DEVELOPMENT of new energy vehicle is an efficient path to energy safety and environmental protection. Permanent magnet synchronous motor (PMSM) is widely used in XEV as the feature of high power density, high efficiency. High speed is one of the developing trend as the need of more higher power density [1]. Meanwhile, the NVH performance of motor is becoming one of the focus to improve the driving comfort of vehicle [2, 3]. However these two kinds of requirement may cause conflict in design. So, rotor design became a focus because it is linked to both electro-magnetic and NVH performance. However, most of papers about rotor topology focused on the electro-magnetic performance.

Cristian Andrei [4] presented improved sinusoidal rotor field poles to reduce the cogging torque and total loss of PMSM for wind turbine application. Through optimizing the air gap ratio of the different rotor shapes, a reduction of the cogging torque of almost 86% and a reduction of the total losses of more than 18% in the rated operation point are achieved comparing to the reference machine to an exemplary adapted geometry.

D. K. Athanasopoulos [5] simulated four permanent magnet synchronous machines with the same stator and different rotor topologies and compared their electromagnetic characteristics.

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The rotor topologies are considered with surface-mounted magnets, magnets embedded in the surface, tangentially embedded magnets and cross buried magnets. A study of the influence of specific rotor design parameters on the PMSM electromagnetic behavior is realized.

Yue Zhang [6] compared three interior PMSM rotor topologies which are U-shaped, V-shaped and $\cap$ -shaped with their parameters including no load magnetic motive force (MMF), torque output under different speed and inner power factor angle. Then V-shaped interior PMSM was chosen as the most suitable EV traction motor scheme mainly according to FEM results of torque and harmonics of MMF.

Arindam Das investigated effects of different magnet configuration, rotor surface profiling and variations in flux guides on important performance parameters like output torque, torque ripple and cogging torque in paper [7]. The four rotor topologies are U-shaped, V-shaped, $\cap$ -shaped and $\cup$ -shaped with mid pole rotor profiling. It is found out that with different magnet configuration, it is possible to obtain a motor model with better performance than the standard Prius 2004 motor.

Xiangdong Liu investigated the parameters influence of four types of rotor topology not only the low-speed region, but also the high-speed region detailedly [8]. Rotor topologies of spoke type, $\cap$ shape, U shape and V shape are quantitatively compared and comprehensive analyzed. At last V shaped rotor was chosen to made and tested.

Xiaoyu Liu presented a general pattern of the PM arrangement in PMSM [9] which can produce different types of PM arrangement in the rotor by varying the parameters of the general pattern. The employed approach combines global optimization method embedded with finite element method (FEM) for solving the optimization of the PM structures. Based on the proposed general pattern and optimization methods, the best PM arrangement in the motor can be automatically determined.

Because of rare earth cost, many new type of rotor topologies are developed which use less or no rare earth magnet. Sinisa Jurkovic and engineers of GM designed two motors for the next generation Chevy Volt vehicle [10]. Machine A is a multi-barrier high reluctance ferrite magnet machine, then machine B is a double-barrier V-shaped PM motor. These two motors redesigned reduced rare earth and heavy rare earth content by over 80% and 50% respectively comparing to the previous generation Chevy Volt while maintaining and improving the drive unit and vehicle performance. The new

designs also bring significant mass saving, 40% for machine B and almost 30% for machine A. However, the authors did not mentioned about the NVH performance much, only torque ripple is mentioned, meanwhile high speed stress of machine A is analyzed except machine B.

J. Juergens presented an innovative air cooled permanent magnet assisted synchronous reluctance machine (PMaSyRM) for A-segment BEV [11]. Low cost ferrite magnets are used in an optimized rotor geometry with high saliency ratio. Chae-Lim Jeong presented a new rotor topology using hybrid type permanent magnet. The capability of resistance to irreversible demagnetization is considered [12].

Besides of EM performance, there are also some papers about the NVH performance related to rotor topology.

For Chevrolet Spark BEV, Sinisa Jurkovic presented another oil-cooled PMAC machine [13]. Based on a double-barrier V-shaped rotor structure, additional slots have been added to the rotor to help shape local saturation paths and in turn lower torque ripple and acoustic noise. The slots are asymmetrically placed from north to south pole, which is an additional key lever for torque ripple and acoustic noise reduction. Finally, the last feature added to combat the torque ripple is rotor surface shaping. The concept involves adding harmonics to the rotor surface, making it uneven circle. At last, tangential and radial force for a speed sweep with peak EM torque are analyzed and shown in two order plots.

Yanqing Chen [14] present a PMSM with U shaped rotor topology. The machine is described based on multi-field coupling theory and machine behavior in aspects of electromagnetic, thermal, vibration and acoustic noise are predicted. Aryanti Kusuma Putri evaluated various approaches to improve the noise, vibration, and harshness (NVH) behavior of a single-layer interior PMSM [15]. At last, the harmonics of the air-gap flux density was reduced through alteration of the rotor surface.

In all, there is few paper about rotor topology research both on the capability to achieve both high speed and low acoustic noise.

This paper presented a solution which considers high speed stress in rotor and low-acoustic noise. Through comparing with V shape rotor and traditional triangle rotor, the advantage of the improved triangle rotor is revealed.

II. INTRODUCTION OF IMPROVED TRIANGLE POLE(ITP) TOPOLOGY

One of difficulties of EM-design of high speed traction motor for EV/HEV lies on the balancing of the high speed, low noise and basic electromagnetic performance.

For design of rotor structure, high speed means better rotor topology on mechanical performance; low noise means low harmonics of magnetic field; basic EM performance need high saliency. So the difficulty of design work will be enhanced greatly when the three kinds of needs reflect on the rotor at the same time.

To solve this problem, the author proposes an improvement scheme based on the "triangle" structure, which can

accommodate the requirements of these three aspects, and the rotor structure is shown in Fig. 1.

The rotor structure has two layers of magnet slot: the first layer which is closer to the air gap as linear shape magnet slot with magnet3 inserted, and the second layer is "V" shape magnet slots which is closer to the inner diameter of rotor with magnet1 and magnet2 inserted. So it combines the characteristics both linear shape rotor structure and V shape rotor structure.

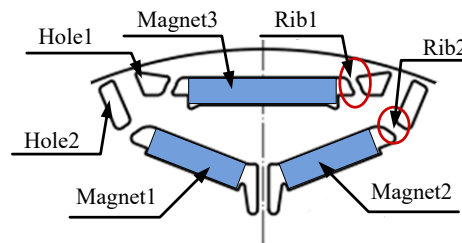


Fig. 1. Improve triangle magnetic pole topology.

On both sides of magnet3, 2 holes (Hole1) are symmetrically arranged, thus formed 2 more ribs (rib1) on both sides of magnet slot in order to constraint the outward movement of magnet3 under high speed. In designing the shape of these two cavities and magnet slot not only relates to the strength of reinforcement ribs, also affect the harmonic distribution of the rotor magnetic field. On both sides of V shape magnet slot, another 2 holes (hole2) are arranged symmetrically, also formed another reinforcement (rib2) to improve the limit speed of the rotor. Two layer structure is not only easy to increase the limit speed of rotor, but also provide effective way to control the harmonic component through adjusting the upper and lower pole-arc coefficient.

In general, the new rotor structure provide a better solution with more balanced performance on higher speed, torque and NVH performance. 4 more ribs are added to reduce the tensile and bending stress of rotor sheet under high speed comparing to traditional triangle rotor topology. And these 4 ribs are all in a convex circular arc form which help to reduce the stress more. Low harmonics in magnetic field make motor silent. The double layer design of permanent magnet is easy to reduce rate of magnetic-field harmonic through adjusting the flux ratio of upper and lower two layers; meanwhile, the four holes added also facilitate to regulate the air-gap flux density waveforms under the no-load and load condition in order to control the total harmonic component. Besides, low harmonic is beneficial to lower iron loss and higher efficient especially under high speed.

III. COMPARISON STUDY OF ITP

The performance advantage of the ITP comparing to traditional V shape pole and triangle pole is studied. The three rotor topologies are shown in Fig.2. For convenience of comparing, stator parameters including air-gap length are all the same, and magnet parameter is set to the same. Key parameters of magnetic poles are shown in Table I.

Obviously, the V shaped pole has the biggest torque and meanwhile the smallest magnet volume. If only torque capability is considered, the V shaped pole will be the best

choice. However, other performance should be considered, for example the maximum speed, the efficiency, the NVH performance.

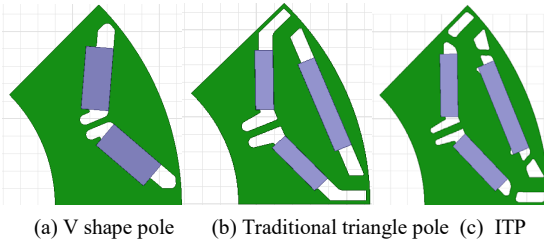


Fig. 2. 3 kinds of magnetic pole for comparing study.

TABLE I
KEY PARAMETERS OF THREE KINDS OF POLE

Parameters	V shaped	traditional triangle	improved triangle
Number of poles/slots	8/48		
Out diameter of rotor /mm	163		
Air-gap length/mm	1		
Stack length/mm	170		
Magnet total area/mm ²	2304	2440	2520
Max torque/Nm	356.0	333.6	312.4
Max phase current/A _{rms}	450		
Max demagnetizing current/A _{rms}	450		

A. Maximum speed

The yield strength of silicon steel material limits the maximum speed. Assuming the yield strength of rotor sheet is 400MPa, the maximum rotation speed of the 3 rotor poles are shown in Table II.

TABLE II
MAXIMUM SPEED* OF THE THREE TYPICAL POLES

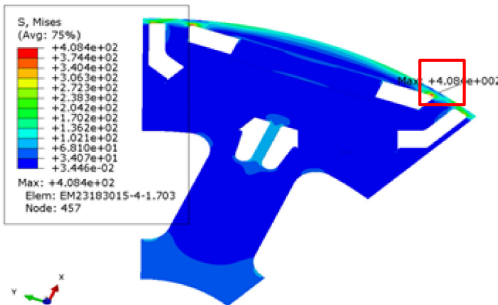
Parameters	V shape pole	traditional triangle pole	improved triangle pole
Max speed/r/min	14417	7921	15326

*Assumption: No glue or encapsulating material in magnet slot

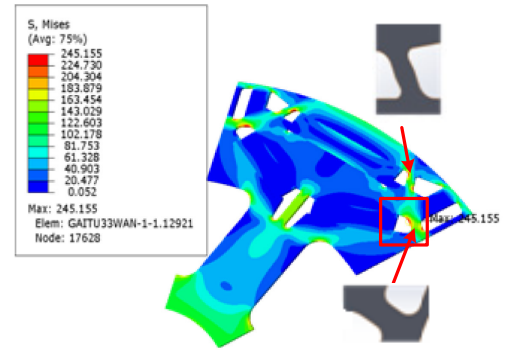
The improved triangle magnet pole has better performance on maximum speed benefited from rib1 and rib2. The two ribs changed the failure place of rotor sheet where maximum stress occurs and decrease the maximum stress at meanwhile. Fig.3 showed the difference of failure place.

B. Air-gap flux density and NVH comparison

Air-gap flux density curve is important to cogging torque, torque ripple, radial force and NVH, so air-gap flux density is studied. The no-load normal flux density waveforms of the three kinds of pole are analyzed in Fig.4. As we can see, the waveforms of two triangle poles are better.



(a) Stress of traditional triangle pole ($n=8000\text{r/min}$)



(b) Stress of ITP ($n=12000\text{r/min}$)

Fig. 3. Stress of rotor sheet.

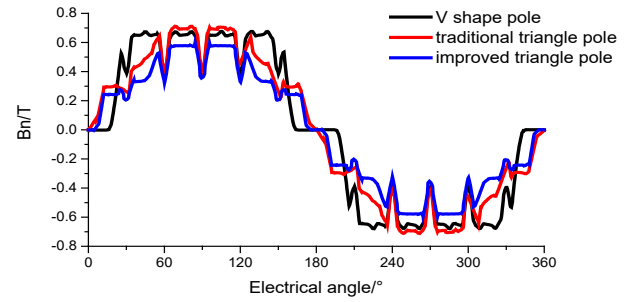


Fig. 4. No-load normal flux density of the 3 poles.

The air-gap flux density of 3 kinds of magnetic pole under maximum torque are shown in Fig.5. Through FFT analysis, we can see the total harmonic distortion of the 3 poles in Table III.

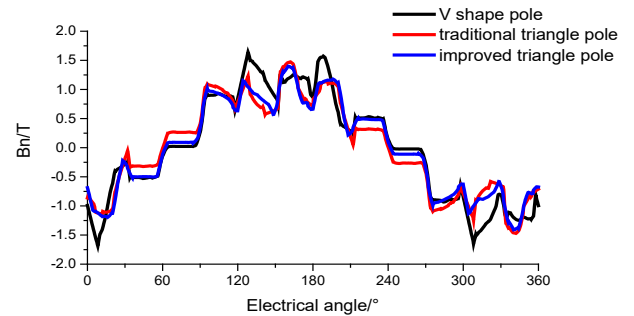


Fig. 5. Normal flux density waveform of 3 kinds of magnetic pole under max torque.

TABLE III
FLUX DENSITY WAVEFORM ANALYSIS OF TOTAL HARMONIC DISTORTION OF THE 3 POLES

Parameters	V shape pole	traditional triangle pole	improved triangle pole
No-load			
Amplitude/T	0.69	0.71	0.59
THD/%	26.15%	15.14%	15.83%
Maximum torque			
Amplitude/T	1.6164	1.4504	1.3998
THD/%	29.66%	28.37%	27.61%

The traditional triangle pole has higher amplitude of no-load flux density and better THD comparing to V shape pole at meanwhile. Through optimization of mechanical performance, the flux density of ITP is decreased because of the increased flux-leakage. And THD at no-load is bigger because of this decrease. However, the THD at maximum torque is better because of the lowest amount harmonics.

From Table III we can easily conclude that the improved

triangle pole has much more sinusoidal flux density waveform than V shape pole whenever no-load or maximum torque. Better NVH performance can be achieved consequently.

For further comparing, the radial force of the 3 poles are analyzed in time domain. The main harmonics are shown in Fig.6. As we can see, most order of radial force is decreased. Better performance of acoustic noise is predicted in Fig.7.

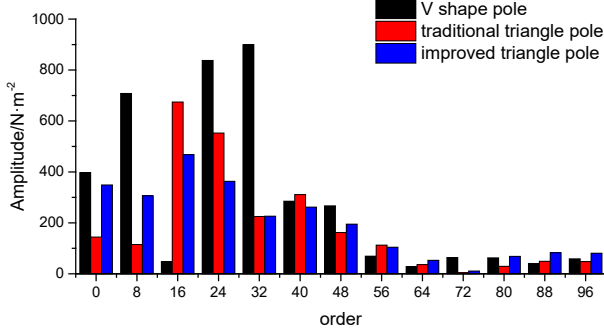


Fig. 6. Main harmonics of radial force under max torque.

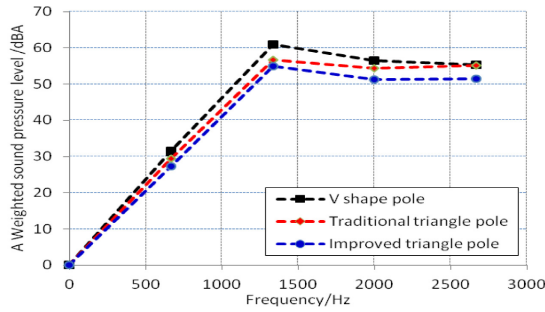
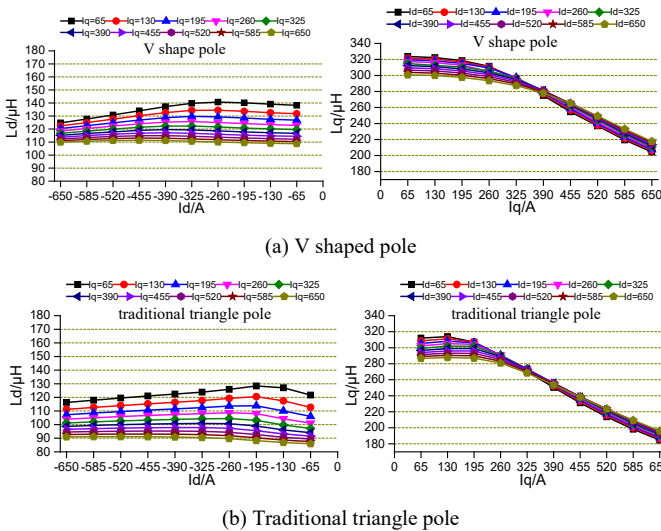


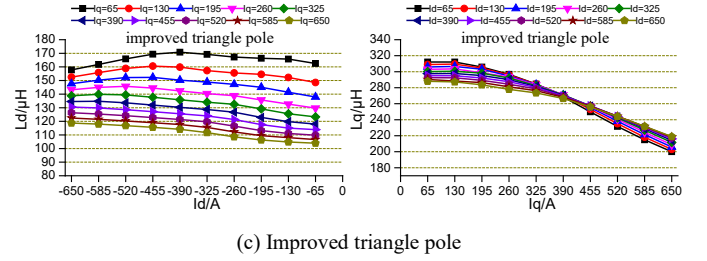
Fig. 7. A weighted sound pressure level under max torque (0.5m sound field).

C. Inductance and torque

Inductance is important to torque capability and power capability, so the inductance of the 3 poles are analyzed in Fig.8. As predicted, L_d and L_q traditional triangle pole is smaller than V shaped pole because of the benefit from pole topology. After optimization of traditional triangle pole topology by adding rib1 and rib2, L_d is increased with the increasing of maximum speed. The L_q is remained almost the same.



(b) Traditional triangle pole



(c) Improved triangle pole

Fig. 8. Inductance of 3 poles.

In order to clearly show the torque capability and proportion of reluctance torque, the torque vs current angle is calculated in Fig.9 and Fig.10. We can easily found that, although the flux and maximum torque was decreased, the proportion of reluctance torque of ITP is still remained at high level.

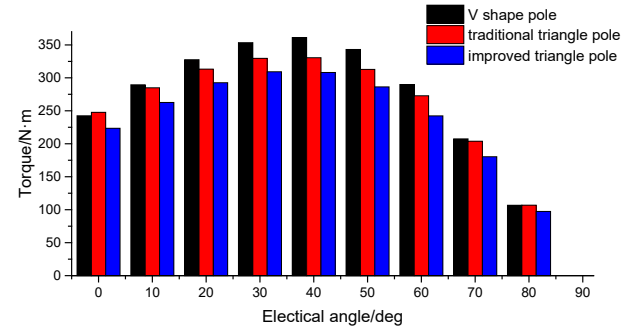


Fig. 9. Torque VS electrical angle ($I=450A_{rms}$).

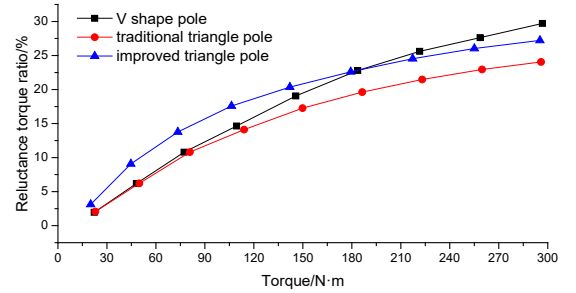


Fig. 10. Reluctance torque ratio.

Another benefit of the ITP is the effect on the torque ripple which is the lowest between 3 poles as shown in Fig.11.

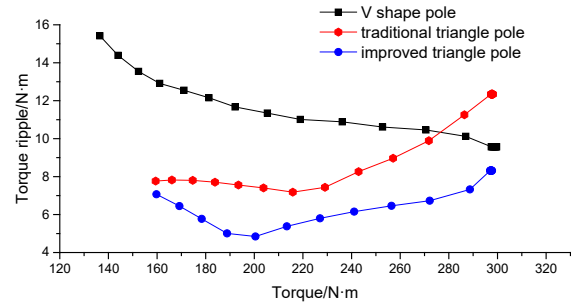


Fig. 11. Torque ripple comparison.

D. Anti-demagnetization capability

Apply maximum demagnetization current to the 3 poles, the lowest working point of magnet is analyzed in Fig.12. As we

can see, the working point of the improved triangle pole is the highest. The magnets in ITP are the safest.

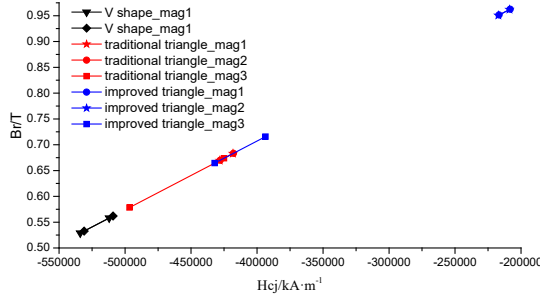
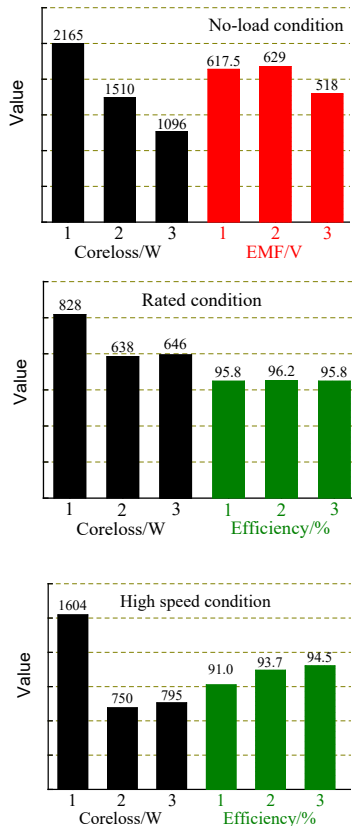


Fig. 12. Demagnetization capability of 3 poles.

E. Efficiency at high speed

Fig.13 is the comparison of efficiency, core-loss under specific working point and EMF. Benefit from the lower flux density on stator and low harmonic advantage, the ITP has better performance on core-loss against V shaped rotor. Meanwhile, the efficiency of ITP at rated condition remain almost the same. Besides, the minimum BEMF and big inductance make the ITP much easier to field-weakening, so the ITP motor has much better efficiency on high speed region consequently.



eg: 1:V shape pole; 2: traditional triangle pole; 3: improved triangle pole.

Fig. 13. Performance comparison.

F. Summary of comparison

Through comparing the max speed, flux density waveform, NVH performance, inductance, torque, and anti-

demagnetization capability, efficiency in Table IV, the presented ITP has many advantages above other kind of poles and more balanced performance.

TABLE IV
QUALITATIVE CONTRAST OF THE THREE KINDS OF POLE

Parameters	V shape pole	traditional triangle pole	improved triangle pole
THD of flux density waveform	△	⊙	⊙
NVH performance	△	⊙	⊙
Maximum speed	○	△	⊙
Saliency	⊙	⊙	○
Anti-demagnetization capability	△	○	⊙
Efficiency at high speed	○	⊙	⊙

△→○→⊙, become better

IV. CONCLUSION

Through research and comparison work on various rotor topologies, the ITP has more balanced performance on many aspects including speed, NVH, reluctance torque component, magnet safety:

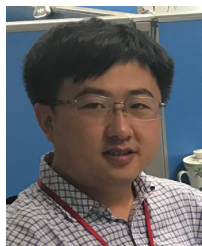
- (1) More higher speed than V shape and triangle shape pole topology.
- (2) Smaller harmonics component than V shape pole, almost the same as traditional triangle shape magnetic pole, and the best NVH performance consequently.
- (3) Although the saliency is decreased, but reluctance torque ratio is remained good especially in high speed.
- (4) Better capability of anti-demagnetization, magnets of this kind of poles is more safer.

In all, the new rotor topology will be a good choice to fulfill the application need of high speed traction motor.

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